Claim-Verified Multi-Hop RAG: Measuring Hallucination Propagation Across Retrieval Steps

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Abstract

Retrieval-augmented generation (RAG) improves factual grounding, but multi-hop RAG introduces a new failure mode in which unsupported intermediate answers can influence later retrieval and generation. We study this risk on HotpotQA bridge questions by comparing three pipelines: single-pass RAG, multi-hop RAG with question decomposition, and verifier-guided multi-hop RAG that checks each hop with a calibrated NLI model and intervenes on unsupported answers. On a shared 2,000-example subset, multi-hop RAG improves Exact Match and strict retrieval recall over single-pass retrieval, confirming the value of hop-wise evidence chaining. Post-hoc grounding analysis shows that unsupported intermediate answers are common and only modestly associated with lower final Exact Match, while many correct answers still contain unsupported hops. Verifier-guided intervention increases hop-level support rates and reduces faithfulness failures among correct answers, with little change in Exact Match relative to unverified multi-hop RAG. These results show that per-hop verification can make intermediate reasoning more evidence-grounded even when end-task accuracy gains are small.

1 Introduction

Retrieval-augmented generation (RAG) has become a standard approach for grounding large language model (LLM) outputs in external evidence, reducing the rate at which models produce factually unsupported claims [1]. By conditioning generation on retrieved passages rather than relying solely on parametric knowledge, RAG systems have demonstrated strong performance on open-domain question answering benchmarks. However, as question complexity increases, single-pass retrieval is often insufficient. In particular, multi-hop questions which require a chain of reasoning across multiple documents have motivated iterative retrieval architectures that decompose a question into sub-questions, retrieve evidence at each step, and pass intermediate answers forward to guide subsequent retrieval [2]. This chaining structure introduces a failure mode that single-hop RAG does not face: error propagation across retrieval steps. If a model produces an unsupported intermediate answer at hop one, that answer may steer hop-two retrieval toward irrelevant passages, producing a final answer that is confident and fluent but entirely unsupported by the retrieved evidence. The challenge is that standard evaluation metrics do not distinguish between these cases. A system can achieve correct final answers for the wrong reasons, or plausible-sounding answers with no grounding in the retrieved context. Existing work on multi-hop QA has focused primarily on accuracy, leaving the faithfulness of intermediate reasoning steps less consistently measured.

This paper studies hallucination propagation across retrieval steps by building and comparing three RAG pipelines of increasing sophistication on HotpotQA [2], restricted to bridge-type questions, those that require finding a bridge entity across two documents and are therefore genuinely dependent on intermediate reasoning. The first pipeline is a basic single-step RAG baseline that uses one hybrid retrieval action before directly generating the answer.The second is a multi-hop RAG pipeline that uses GPT-4o-mini to decompose the input question into sub-questions, retrieves evidence independently at each hop, and chains intermediate answers into the final generation. The third pipeline adds verifier-guided intervention. Each hop answer is converted into a question-answer claim and checked against the top retrieved passages with a calibrated NLI verifier. When a hop is flagged as unsupported, the system runs a targeted rescue step that re-retrieves with a higher k and regenerates before continuing.

We test three hypotheses.
H1: Multi-hop decomposition improves strict recall and EM relative to single-pass RAG.
H2: Unsupported hop-1 answers are associated with lower final-answer EM, indicating hallucination propagation.
H3: Verifier-guided intervention on unsupported hops reduces ungrounded intermediate answers and improves final EM relative to unverified multi-hop RAG.

We evaluate the three pipelines on a shared 2,000-

question HotpotQA bridge subset and compare both accuracy and hop-level grounding. Our goal is to study whether unsupported intermediate claims compound into incorrect final answers, and whether per-hop verification can reduce this failure mode by increasing the rate of evidence-supported intermediate answers.

2. Related Work

2.1 Retrieval-Augmented Generation

Lewis et al. (2020) introduced the RAG framework, combining a parametric language model with a non-parametric dense retrieval component to ground generation in external documents [1]. Subsequent work on Dense Passage Retrieval showed that bi-encoder models, when trained using in-batch negatives, significantly outperform the BM25 baseline for open-domain QA retrieval. In this setup, the bi-encoder maps questions and source passages into separate vector embeddings, while the in-batch negative framework uses concurrent data blocks to train the model on correct matches and negative distractors simultaneously [3]. More recent work has shifted toward hybrid retrieval by combining keyword-based search with meaning-based semantic search, finding that neither approach dominates across all question types, motivating fusion strategies such as Reciprocal Rank Fusion [4].

2.2 Multi-Hop Question Answering

HotpotQA [2] established the bridge-question setting as a benchmark for multi-hop reasoning, requiring systems to chain evidence across two Wikipedia documents. Subsequent multi-hop datasets such as 2WikiMultiHopQA and MuSiQue introduced harder chaining settings, but we focus on HotpotQA bridge questions to isolate hop-propagation failures [5][6].On the systems side, iterative retrieval approaches such as IRCoT interleave retrieval with chain-of-thought generation, using each reasoning step to issue a new retrieval query [7]. FLARE similarly triggers retrieval dynamically based on model uncertainty during generation [8]. These methods improve answer accuracy on multi-hop benchmarks but do not explicitly verify whether intermediate answers are supported by retrieved evidence before using them to guide subsequent retrieval.

2.3 Faithfulness and Hallucination in RAG

A growing body of work distinguishes answer correctness from answer faithfulness, which is whether a generated answer is actually supported by the provided context. Yoran et al. (2023) showed that RAG systems remain brittle to irrelevant retrieved passages, often generating plausible but unsupported answers [9]. Min et al. (2023) proposed FActScore to decompose long-form generations into atomic claims and verify each against a knowledge source [10]. Honovich et al. (2022) explored using NLI models as automatic verifiers of statement-to-context entailment, finding strong correlation with human faithfulness judgments [11]. This paper extends these ideas to the multi-hop setting, where faithfulness failures at intermediate reasoning steps have the potential to compound across hops rather than appearing in isolation.

3. Methodology

3.1 Dataset

We use the HotpotQA distractor setting [2], which provides 10 candidate paragraphs per question which contains 2 gold supporting paragraphs and 8 distractors, making it a realistic retrieval benchmark rather than a reading comprehension task over pre-selected gold passages. We restrict evaluation to bridge questions, which require finding an intermediate entity in one document to locate the answer in a second document. Comparison questions, which involve parallel lookups of two independent facts, are excluded because their two retrieval steps are independent and do not exhibit the hop-propagation failure mode under study. This yields 5,918 bridge questions in the validation set. While Pipeline 1 is evaluated on the full set, the multi-hop pipelines require several LLM calls per question, so all three pipelines are compared on a shared, representativeness-validated subset of 2,000 bridge questions; the sampling procedure and comparison controls are detailed in Section 3.5.

3.2 Hybrid Retrieval

Each question in the data set comes packaged with its own 10 paragraphs, many of which are the same Wikipedia articles appearing as either the gold standard or distractors across different questions. We build a single flattened corpus by pulling out all these paragraphs into a single list and de-duplicating all paragraphs from the validation set, yielding 54,391 unique passages. Without de-duplication there would be many identical copies in the corpus which would not accurately simulate searching a real knowledge base, and for dense retrieval it would result in duplicated vectors in FAISS. Retrieval combines BM25 and dense FAISS (all-MiniLM-L6-v2) rankings with Reciprocal Rank Fusion (RRF), then returns the top-k passages (k=10, selected via ablation; see Appendix A). This shared retriever is used identically across all three pipelines so accuracy differences reflect pipeline architecture rather than retrieval tuning.

3.3 Evaluation Metrics

All pipelines are evaluated with the following metrics:
Exact Match (EM): 1 if the normalized prediction equals the gold answer following HotpotQA/SQuAD normalization. This is the primary accuracy signal and the main basis for comparing pipelines.
Token-F1: Token-level F1 between normalized prediction and gold gives partial credit when answers are multi-word and partially correct.
Retrieval Recall@k (soft): Fraction of examples with at least one gold paragraph in the top-k retrieved set. Measures whether any relevant evidence was found.
Retrieval Recall@k (strict): Fraction of examples with both gold paragraphs in the top-k set. Measures whether both required documents were found, a stricter criterion for bridge questions than soft recall.
Abstention rate: Fraction of examples whose final prediction is UNANSWERABLE. A high abstention rate indicates the pipeline is failing to ground answers in retrieved context rather than generating unsupported responses.

Pipelines 2 and 3 also report the following grounding metrics:
Per-hop verification failure rate : Fraction of hop-i answers labeled UNSUPPORTED by the NLI verifier against that hop's retrieved passages. Measures how often intermediate claims are ungrounded and provides the hop labels used by the three metrics below.
Conditional EM by grounding state: Mean final EM within three mutually exclusive groups defined by per-hop verification labels: both hops grounded, one hop flagged, and both hops flagged. This tests if unsupported intermediate claims propagate into worse final answers.
Faithfulness failure rate: Among examples with final EM = 1, the fraction with at least one ungrounded hop. This captures the "right answer for the wrong reason" case: a correct final answer that is not supported by retrieved evidence. Standard accuracy metrics count these as successes, so this metric surfaces a faithfulness failure that EM alone misses.
Propagation penalty: Difference in mean final EM between examples whose hop-1 answer is grounded and those whose hop-1 answer is ungrounded. Estimates the accuracy cost of propagating an unsupported intermediate answer into later hops. A large penalty supports the claim that per-hop faithfulness matters for multi-hop QA accuracy.

3.4 Verifier Calibration

Verifier calibration was performed post-hoc on saved Pipeline 2 outputs to ensure that hop-level grounding labels were meaningful before use in Pipeline 3 intervention. Each hop-1 answer was converted from a question-answer pair into a declarative claim and verified against retrieved evidence using local NLI models. Entailment was scored per passage (top 3 retrieved passages) and aggregated by maximum entailment. A hop was labeled supported when at least one passage exceeded an entailment threshold of 0.50, selected via threshold ablation in Appendix B. We compared three candidate NLI verifiers and tracked UNANSWERABLE as a separate state. The highest-lift model, cross-encoder/nli-deberta-v3-base, was selected for full two-hop grounding analysis and for Pipeline 3 intervention.

3.5 Shared Evaluation Subset and Comparison Controls

Because the multi-hop pipelines issue three to four LLM calls per question (decomposition, per-hop generation, and final synthesis), evaluating all three pipelines on the full 5,918-question bridge set would require roughly three days of sequential API calls under the provider's rate limits. We therefore evaluate all pipelines on a shared random subset of 2,000 bridge questions (seed=42), drawn from the completed Pipeline 1 full-set results. Prior to committing API budget, we validated that this subset is representative of the full set by requiring the subset Exact Match to fall within two points of the full-set value and to lie within a bootstrap 95% confidence interval; this criterion was met, and the full comparison is reported in Table 1 (Section 5.1).

To ensure that observed differences across pipelines reflect architecture rather than confounds, all pipelines are evaluated on this identical subset with the same fixed hybrid retriever (BM25 + dense FAISS with RRF fusion, k=10). We use GPT-4o-mini for question decomposition and answer generation across all pipelines because it offers strong short-form QA performance at a cost that makes large-scale multi-hop evaluation with repeated retrieval and regeneration steps feasible. All pipelines use GPT-4o-mini with context-only answering, UNANSWERABLE abstention, and temperature 0. Pipelines 2 and 3 reuse the same saved question decompositions so retrieval queries start from identical sub-questions.

4. Pipeline Designs

4.1 Pipeline 1: Naive Single-Pass RAG

Pipeline 1 establishes the baseline performance for this study using a naive single-pass retrieval-augmented generation (RAG) design. For each HotpotQA bridge question, the system performs one retrieval step over a shared Wikipedia paragraph corpus and then generates a final answer directly from the retrieved context, without intermediate decomposition or hop-level verification. This baseline is critical because it quantifies how far a standard one-shot RAG setup can go before introducing multi-hop reasoning or verifier-based controls.

4.2 Pipeline 2: Multi-Hop RAG

Pipeline 2 is a multi-hop retrieval-augmented generation (RAG) pipeline designed to improve over single-pass QA by explicitly chaining intermediate reasoning steps. For each HotpotQA bridge question, the system decomposes the original query into two sequential sub-questions (Q1, Q2) using gpt-4o-mini, retrieves evidence at each hop with a fixed hybrid retriever, and then generates a final answer from the combined hop contexts. Hop 2 retrieval is query-augmented with the hop 1 answer unless hop 1 returns UNANSWERABLE, in which case hop 2 falls back to the unaugmented Q2 query. This makes Pipeline 2 the "unverified propagation" baseline: intermediate answers are propagated forward without a verifier-mediated intervention. Grounding of each intermediate answer is assessed post-hoc using the verifier described in Section 3.5, so that faithfulness can be measured without altering Pipeline 2's generation behavior.

4.3 Pipeline 3: Claim-Verified Multi-Hop RAG

Pipeline 3 starts from the same multi-hop architecture as Pipeline 2, then adds per-hop verification and intervention. After each hop answer is generated, it is converted into a question-answer claim and checked against the top retrieved passages with the calibrated NLI verifier from Section 3.4. The primary policy intervenes only on UNSUPPORTED hops by re-retrieving with a higher k, regenerating with a stricter prompt, and re-verifying once before continuing. UNANSWERABLE hops are recorded but not rescued in the main comparison.

5. Results

5.1 Pipeline 1 Results: Naive Single-Pass RAG

The baseline retrieves the top-10 passages for the input question in a single step and generates an answer directly from the concatenated context. Table 1 reports results for the naive single-pass baseline on both the full bridge-question validation set and the shared evaluation subset.

Table 1 confirms this subset is representative of the full bridge set: subset EM (0.3385, 95% CI: 0.318–0.361) is within 0.18 points of full-set EM (0.3403) and contains it in the confidence interval, with all other metrics differing by less than one point. Subsequent cross-pipeline comparisons therefore use the subset column as the Pipeline 1 reference.

Table 1. Pipeline 1 performance on the full bridge set and the shared evaluation subset.

Metric	Full set (n=5,918)	Subset (n=2,000)
EM	0.3403	0.3385
F1	0.4459	0.4532
Recall@k (soft)	0.9542	0.9525
Recall@k (strict)	0.4633	0.4610
Abstention rate	0.3552	0.3470

On the subset, the baseline reaches EM 0.3385 and F1 0.4532. Soft recall is high (0.9525), but strict recall is only 0.4610, so more than half of questions miss at least one gold paragraph. This soft–strict gap is predominantly driven by bridge questions, where the second document can only be identified after reading the first, and it represents the core failure mode that single-pass retrieval cannot resolve regardless of k. Because bridge questions require chaining two facts, this strict-recall bottleneck caps the achievable answer accuracy of any one-pass pipeline. Abstention is also high at 0.3470, so the model declines roughly one third of questions when the retrieved context is judged insufficient.

5.2 Pipeline 2 Results: Multi-Hop RAG

Decomposing each question into two sub-questions and chaining retrieval across hops produces large gains over the single-pass baseline on every accuracy metric. Table 3 reports the full three-pipeline comparison on the shared 2,000-question subset and Pipeline 1 vs Pipeline 2 differences are also summarized in Appendix C. The largest gain is in strict recall, which rises from 0.4610 to 0.6950. This validates the rationale for the multi-hop approach, as using the bridge entity from hop 1 to refine the hop 2 query retrieves crucial evidence missed by single-pass retrieval. The improved evidence chain translates directly into answer quality, with EM increasing by 8.55 points and F1 by 10.19 points. Abstention falls from 0.3470 to 0.2400, indicating that the model more often judges the retrieved context sufficient to answer when both supporting documents are present.

Accuracy gains do not imply faithful intermediate reasoning. Using the calibrated verifier from Section 3.4, each intermediate answer was checked post-hoc against its retrieved passages. Hop 1 was labeled Unsupported in 32.7% of examples and 32.9% in hop 2, while Unanswerable was tracked as a separate state. Table 2 reports conditional EM on decided cases only, where both hops are either Supported or Unsupported.

Table 2: Pipeline 2 Conditional EM by grounding state (decided-only; n=1,126).

Grounding state	n	Mean EM
Both hops supported	336	0.5982
One hop unsupported (combined)	568	0.5123
Hop 1 only unsupported	232	0.5043
Hop 2 only unsupported	336	0.5179
Both hops unsupported	222	0.5495

Conditional Exact Match (EM) scores are highest when both hops are supported (0.5982, n = 336). Performance decreases when exactly one hop is unsupported (0.5123, n = 568), showing similar drops whether only hop 1 is unsupported (0.5043, n = 232) or only hop 2 is unsupported (0.5179, n = 336). However, error rates do not follow a strictly monotonic pattern. When both hops are unsupported, the EM score actually rises to 0.5495 (n = 222), outperforming the single-hop unsupported group. This indicates that while unsupported intermediate claims occur frequently, they do not translate into final errors through a simple, compounding stepwise process. This non-monotonic pattern may also reflect residual verifier noise, suggesting that stronger calibration or more precise claim verification could further improve hop-level grounding labels. Furthermore, the higher performance in the both hops unsupported group likely points to a fallback to parametric memory, where the model relies on internal priors when it detects that the retrieved evidence is entirely unreliable.

To test H2, we examined the association between hop-1 grounding and final accuracy to determine if unsupported intermediate claims propagate through the reasoning chain and compound into incorrect final answers Surprisingly, the propagation penalty was limited to +0.0313, with mean EM dropping only from 0.5076 to 0.4763 when the first hop lacked evidence. This finding provides only partial support for H2, although unsupported early-stage answers do correlate with lower final EM, the marginal scale of the drop suggests that hallucination compounding is a secondary risk rather than the primary driver of downstream failure. Crucially, 60.9% of all correct final answers still contained at least one unsupported hop, demonstrating a frequent tendency to generate the right answer for the wrong reason (Table 4). Overall, Pipeline 2 improves answer quality over single-pass RAG but still leaves a large gap between correctness and evidence-grounded reasoning. Pipeline 3 tests whether verifier-guided intervention can reduce unsupported intermediate hops without sacrificing the accuracy gains from multi-hop retrieval.

5.3 Pipeline 3: Claim-Verified Multi-Hop RAG

Pipeline 3 uses the same multi-hop architecture as Pipeline 2, then adds per-hop verification and a rescue step for Unsupported hops. Table 3 reports standard QA metrics for all three pipelines on the shared subset. Table 4 reports grounding deltas relative to Pipeline 2.

Table 3: Pipeline comparison on the shared evaluation subset (n=2,000, seed=42).

Metric	P1	P2	P3	Δ
EM	0.3385	0.4240	0.4250	+0.0010
F1	0.4532	0.5552	0.5538	−0.0013
Recall@k (soft)	0.9525	0.9570	0.9560	−0.0010
Recall@k (strict)	0.4610	0.6950	0.6810	−0.0140
Abstention rate	0.3470	0.2400	0.2495	+0.0095

Δ = P3 − P2

Relative to Pipeline 2, Pipeline 3 leaves end-task accuracy essentially unchanged. EM rises by only 0.10 points, F1 falls by 0.13 points, and abstention rises slightly. Strict recall declines by 1.4 points. The intervention is therefore not a large accuracy win under standard QA metrics.

The main effect appears in faithfulness. Intervention was triggered on 33.1% of hop-1 answers and 33.2% of hop-2 answers, with rescue accepted on 10.8% and 7.8% of hops, respectively.

Table 4: Pipeline 2 vs Pipeline 3 grounding summary (n=2,000).

Metric	P2	P3	Δ
Hop 1 unsupported rate	0.3265	0.2235	−0.1030
Hop 2 unsupported rate	0.3290	0.2550	−0.0740
Faithfulness failure rate	0.6085	0.4965	−0.1120
Propagation penalty	0.0313	−0.0190	−0.0503
EM | hop 1 supported	0.5076	0.4933	−0.0142
EM | hop 1 unsupported	0.4763	0.5123	+0.0360

Δ = P3 − P2; propagation penalty = EM|sup − EM|unsup

Pipeline 3 reduces hop-1 unsupported rate by 10.3 points and hop-2 unsupported rate by 7.4 points. Among correct final answers, faithfulness failure falls from 60.9% to 49.7%. These results provide mixed support for H3: while the verifier-guided intervention successfully reduced hop-level unsupported rates and mitigated faithfulness failures among correct answers, these grounding improvements did not translate into the predicted lift in final-answer EM. The propagation-penalty row should be interpreted cautiously. The propagation penalty proves unstable, reversing from a marginal +0.0313 in Pipeline 2 to -0.0190 in Pipeline 3, suggesting that hop-1 grounding status is unstable and not a reliable predictor of final correctness. The clearer Pipeline 3 gains are the reductions in unsupported rates and faithfulness failure, not a stronger monotonic compounding pattern.

6. Discussion

Our results show a clear divergence between answer correctness and intermediate-step faithfulness. Multi-hop decomposition (Pipeline 2) strongly supports H1, substantially increasing strict retrieval recall and significantly raising EM by enabling the recovery of bridge entities that single-pass RAG misses. However, this accuracy gain masks a hidden faithfulness risk. Post-hoc analysis only partially supports H2: while a propagation penalty exists, unsupported intermediate claims do not always lead to final errors, often resulting in correct answers that are not traceable to retrieved evidence.

Verifier-guided intervention (Pipeline 3) provides targeted support for H3 regarding faithfulness, though not final accuracy. While Exact Match remained flat, the intervention significantly reduced hop-level unsupported rates and lowered the share of “unfaithful” correct answers. This suggests that per-hop verification is a robust reliability mechanism for improving grounding quality, even if it does not immediately improve headline QA scores. The non-monotonic EM patterns observed in our conditional grounding analysis likely reflect residual verifier noise and the model’s ability to rely on parametric knowledge when retrieval fails.

In high-stakes domains where a correct answer with ungrounded reasoning is an unacceptable risk, verifier-guided architectures may provide the reasoning traceability and structural accountability necessary to ensure that complex multi-hop answers are demonstrably anchored in evidence rather than mere model intuition. Future work should explore stronger, task-specific verifiers and richer intervention policies like verifier-informed reranking, ideally across broader multi-hop datasets, to improve both faithfulness and Exact Match.

7. Conclusion

We evaluated three RAG pipelines for multi-hop QA, moving from single-pass retrieval to multi-hop decomposition and then to verifier-guided intervention. Multi-hop retrieval clearly improves answer quality over single-pass RAG, especially on strict recall and EM, confirming the benefit of hop-wise evidence chaining on bridge questions. However, post-hoc grounding analysis shows that unsupported intermediate reasoning remains common and that many correct answers are still not fully evidence-grounded. Verifier-guided intervention successfully shifts toward higher reliability by significantly reducing ungrounded reasoning steps and faithfulness failures compared to the unverified multi-hop baseline. It demonstrates that per-hop verification is a valuable reliability mechanism even when end-task accuracy gains are small. Overall, the results support a key practical takeaway that evaluating only final-answer correctness is insufficient in multi-hop RAG. Reliable systems should measure and control intermediate grounding explicitly, since correctness and faithfulness can diverge in both baseline and intervention settings. Using a local NLI model rather than a generative LLM-as-a-judge can provide a more cost-effective and scalable verification layer, while its narrow, discriminative task design reduces the risk of the verifier hallucinating its own justifications.

References

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela, 2020, Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.

[2] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning, 2018, HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering.

[3] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih, 2020, Dense Passage Retrieval for Open-Domain Question Answering.

[4] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher, 2009, Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods.

[5] Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa, 2020, Constructing A Multi-hop QA Dataset for Comprehensive Evaluation of Reasoning Steps.

[6] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal, 2022, MuSiQue: Multihop Questions via Single-hop Question Composition.

[7] Harsh Trivedi, Niranjan Balasubramanian, and others, 2022, Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions (IRCoT).

[8] Zhengbao Jiang, Frank F. Xu, and others, 2023, Active Retrieval Augmented Generation (FLARE).

[9] Ori Yoran, Tomer Wolfson, Ori Ram, Jonathan Berant, 2023, Making Retrieval-Augmented Language Models Robust to Irrelevant Context.

[10] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, and others, 2023, FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation.

[11] Or Honovich, Roee Aharoni, Jonathan Herzigand others, 2022, TRUE: Re-Evaluating Factual Consistency Evaluation.

Appendix

Appendix A: Retrieval Configuration Ablation

We sweep retrieval mode (BM25-only, dense-only, hybrid) and top-k (2, 5, 10) on a 200-example random subset prior to full evaluation. For hybrid retrieval, each retriever returns 100 candidate passages. are merged using Reciprocal Rank Fusion (RRF):

$$\text{Score}(d) = \sum_i \frac{1}{k_{rrf} + r_i(d)} \tag{1}$$

where $r_i(d)$ is the rank of document d in list i, and $k_{rrf} = 60$ is the standard smoothing constant. The top-k fused passages are returned as context.

Table A: Retrieval Ablation Results (n=200, seed=42).

Mode	k	EM	F1	Soft Recall@k	Strict Recall@k	Abstention
BM25	2	0.245	0.310	0.750	0.120	0.550
BM25	5	0.300	0.381	0.850	0.235	0.445
BM25	10	0.315	0.412	0.880	0.330	0.415
Dense	2	0.255	0.332	0.830	0.165	0.480
Dense	5	0.325	0.420	0.905	0.330	0.420
Dense	10	0.320	0.430	0.935	0.405	0.370
Hybrid	2	0.275	0.334	0.825	0.130	0.510
Hybrid	5	0.330	0.417	0.910	0.315	0.390
Hybrid	10	0.365	0.456	0.965	0.450	0.350

Appendix B: Verifier Calibration

Before using hop-level grounding labels for analysis and Pipeline 3 intervention, we calibrated the verifier on saved Pipeline 2 outputs. The goal was to check whether NLI flags meaningfully separate higher-risk and lower-risk intermediate answers.

B.1 Setup

We used hop-1 answers from the shared 2,000-example evaluation subset. For each example, the hop question and generated answer were converted into a declarative claim. The claim was then checked against retrieved evidence using a local NLI model. To better match retrieval structure, we did not concatenate all passages into one premise. Instead, we scored entailment for each of the top 3 retrieved passages and took the maximum entailment score. A hop was labeled:

- **SUPPORTED** if $e_{max} \geq \tau$
- **UNSUPPORTED** otherwise
- **UNANSWERABLE** if the model abstained (tracked separately)

B.2 Selection Criterion

Candidate verifiers were compared by lift:

$$\text{Lift} = \text{Fail} \mid \text{UNSUPPORTED} - \text{Fail} \mid \text{SUPPORTED} \tag{2}$$

where failure means final Exact Match = 0. Higher lift means unsupported labels better identify cases that end in incorrect final answers. UNANSWERABLE cases were excluded from this decided-support comparison so abstention was not mixed with unsupported generation.

B.3 Model Ablation

We compared three off-the-shelf NLI models on hop-1 outputs:

Table B3. Verifier Model Ablation (hop 1; n=2,000)

Model	**Unsup %**	**Unans %**	**Fail|Unsup**	**Fail|Sup**	**Lift**	**n**
`nli-deberta-v3-base`	32.6%	21.2%	52.4%	49.2%	+0.031	1575
`DeBERTa-v3-base-mnli`	22.0%	21.2%	48.9%	51.2%	−0.023	1575
`DeBERTa-v3-large-mnli`	21.1%	21.2%	51.2%	50.3%	+0.009	1575

The cross-encoder/nli-deberta-v3-base verifier had the highest lift and was selected for all subsequent grounding analysis and Pipeline 3. Lift is modest, so the verifier is a useful risk signal rather than a perfect detector.

B.4 Threshold Ablation

Using the selected model, we swept entailment thresholds $\tau \in \{0.30, 0.40, 0.50, 0.60, 0.70\}$

Table B4. Threshold Sweep (hop 1; n=2,000).

τ	**Unsup %**	**Unans %**	**Fail|Unsup**	**Fail|Sup**	**Lift**	**n**
0.30	71.2%	21.2%	51.2%	44.0%	+0.072	1575
0.40	72.0%	21.2%	51.1%	44.4%	+0.067	1575
0.50	72.5%	21.2%	51.1%	43.5%	+0.076	1575
0.60	73.0%	21.2%	51.1%	44.0%	+0.071	1575
0.70	73.5%	21.2%	50.9%	45.3%	+0.056	1575

The highest-lift threshold was $\tau = 0.50$. We therefore used this threshold for full two-hop grounding analysis and for Pipeline 3 intervention.

B.5 Summary

Calibration selected cross-encoder/nli-deberta-v3-base with $\tau = 0.50$. This setting was then applied consistently for Pipeline 2 post-hoc grounding diagnostics and Pipeline 3 verifier-guided intervention.

Appendix C: Pipeline 1 vs Pipeline 2 Accuracy Comparison

Table C1 reports pairwise accuracy metrics for Pipeline 1 and Pipeline 2 on the shared evaluation subset. The full three-pipeline comparison appears in Table 3 in the main text.

Table C1. Pipeline 1 vs Pipeline 2 Comparison (n=2,000, seed=42)

Metric	**Pipeline 1**	**Pipeline 2**	
EM	0.3385	0.4240	+0.0855
F1	0.4532	0.5552	+0.1019
Recall@k (soft)	0.9525	0.9570	+0.0045
Recall@k (strict)	0.4610	0.6950	+0.2340
Abstention rate	0.3470	0.2400	0.1070

The largest gain is in strict recall (+0.2340), with corresponding improvements in EM and F1 and a decrease in abstention. Soft recall remains nearly unchanged.